AI Applied to Football



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Knowledge Informed Playing Style Modelling and Explainable Manager Player Compatibility

MIAD (Máster en Inteligencia Artificial Aplicada al Deporte)

Final Project Memoria

August 2026

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This report is structured according to the programme's required three level format, described in `Guidelines/Estructura de documentacion TFM.pdf`, with the technical development organised around CRISP DM, described in `Guidelines/MIAD TFM_ Exemplos propostas final project.pdf`. Every fact, number, and claim in this document is explained in full inside this document itself, not left as a pointer to another file. Supporting files inside this repository, the decision log, the data dictionary, the notebooks, the audits, and the qualitative research documents, are cited so that a reader can go and verify a claim independently if they wish, but the substance of what those files found is always explained here first, in plain language, before the citation.

ESCALON 1, ENFOQUE DIRECTIVO

1. Resumen Ejecutivo

This project asked a single question. Can football domain knowledge and player performance data be combined with machine learning to model the playing philosophies of three specific, real, contemporary managers, Pep Guardiola, Jurgen Klopp, and Roberto De Zerbi, and can that combined model estimate, in a way a football person could actually read and check, how well an individual player fits each manager's footballing identity.

The approach taken was knowledge informed by design rather than purely data driven. Instead of feeding raw statistics into a black box model and hoping something useful came out, a football knowledge layer was built explicitly, by hand, and argued on football grounds at every step. Raw match level numbers, things like the number of passes a player completed into the final third, were first mapped onto ten named football concepts such as progressive passing, pressing intensity,

and box presence. Those ten concepts were then combined into seven broader tactical dimensions, using a set of weights chosen and justified on footballing grounds rather than fitted purely to whatever combination produced the best looking number. The seven dimensions themselves were not invented from nothing. They were anchored on a published, peer reviewed, statistically validated taxonomy of football playing styles, described in full in section three of this report, so that the project's own feature set carries external legitimacy that does not depend on this project's own results being good. Every player and every manager ends up represented as a vector of scores across these seven dimensions, and compatibility between a player and a manager is computed as the cosine similarity between their two vectors. This choice matters and is explained in detail in section 6.4, but the short version is that a similarity computed between two named, human readable vectors can always be decomposed back into which specific football characteristics drove it, whereas a black box classifier cannot, and a black box classifier trained directly on the label of which manager a player actually played under would risk simply learning to recognise the player's club rather than the manager's footballing philosophy, which is precisely the confound this whole project set out to separate from a genuine signal.

The single most important result of this project is not one number. It is a sequence of numbers that tells an honest story about how the evaluation phase actually unfolded, because the sequence includes two real mistakes that were caught before they were reported. The very first working version of the compatibility model produced a strikingly clean pattern that appeared, on the surface, to show a large and exciting difference between the three managers. Rather than accepting this pattern as a genuine finding, it was investigated on the basis that a result this clean, this early, in a domain this noisy, was more likely to be an artefact than a discovery. That investigation found two real, non trivial problems in the data pipeline, described fully in section 6.5, and once both were fixed, the exciting pattern disappeared completely. The corrected model, using only the manager attributed sample that was available at that point, matched a player to the manager he actually played under thirty two point eight percent of the time, restricted to comparisons where enough data was genuinely available, a figure that is statistically no better than pure chance among three managers. This was a disappointing result, and it was reported as disappointing, because reporting an honest negative finding is more valuable to this field than reporting a false positive.

The story did not end there. Two further rounds of work, each one tested against the data rather than simply assumed to help, materially improved this result. Recovering additional season level data for two leagues that had been missing from the original match level source raised the figure to just under thirty eight percent. A subsequent review of the whole pipeline, conducted specifically from the perspective of a football analyst rather than an engineer, identified three concrete, football grounded weaknesses in how players were grouped by position, how pressing statistics were measured, and how goalkeeper behaviour was represented. Fixing all three raised the final figure to forty two point one percent, a real and material improvement over chance, not statistical noise, confirmed by a formal significance test described in section 6.5 that gives a ninety five percent confidence interval of thirty seven point eight to forty six point two percent and a permutation test p value of zero point zero zero zero two against an empirical chance baseline of thirty four percent. Pressing intensity, once properly adjusted for how much of the ball a team's opponents

actually had, turned out to be by a wide margin the single most informative tactical dimension in the whole feature set, and this project can say with real statistical confidence that it separates these three managers more clearly than any other measurable characteristic used here. A closer look at exactly what pressing intensity shows, described fully in section 8.3, produced one of the most interesting and most honestly reported findings in the whole project. Roberto De Zerbi's teams do press distinctly less than the other two managers' teams, exactly as this project's own qualitative research into his tactics predicted in advance. But the popular idea that Jurgen Klopp's teams press harder than Pep Guardiola's does not hold up in the final, corrected data. Guardiola's teams actually score marginally higher on the possession adjusted pressing measure than Klopp's do, the opposite of what the popular narrative around these two managers would suggest, and the most likely explanation, discussed at length in section 8.3, is itself a further demonstration of the exact confound this project has been honest about throughout, squad and league strength still leaking into even the most carefully adjusted measures.

Beyond the quantitative pipeline, this project also explored, and reported descriptively rather than quantitatively, a further signal drawn from each team's starting formation in every match, described fully in section 6.6. That exploration surfaced a genuine data availability complication, also described in section 6.6, which is itself a small but real addition to this project's own record of things that can quietly go wrong in a data pipeline if you are not paying attention.

This project therefore delivers four things of real value. First, a complete, reproducible, documented pipeline covering both match level and season level football data, hardened against two genuine bugs that were found and fixed rather than hidden. Second, a sourced, literature grounded qualitative account of three managers' footballing philosophies that stands on its own regardless of what any quantitative model finds. Third, an honest, carefully diagnosed account of exactly why a plausible, well built, knowledge informed approach did not clear the bar it would need to clear to be called a decisive success, together with a specific, testable explanation for why. Fourth, a fully working interactive application that lets anyone search any player from the top five European leagues, see their compatibility with all three managers, and read exactly why, with the project's own limitations displayed prominently inside the application itself, not hidden in a report nobody will read before using the tool.

2. Introduccion (Vision Macro)

2.1 The problem and its relevance to the sports sector

Football clubs routinely face a decision with large financial and sporting consequences. Will a given player's individual characteristics actually fit a particular manager's system. A club can sign a technically excellent, expensive player who simply does not suit how the team is meant to play, and the failure has little to do with talent and everything to do with tactical fit, which is notoriously hard to measure. Recruitment departments today rely heavily on video analysis, scouting reports, and the accumulated intuition of experienced staff, all of which is genuinely valuable but is also slow, subjective, and difficult to scale consistently across the very large number of players a club

might realistically consider in a single transfer window.

At the same time, football has become one of the most heavily instrumented sports in the world. Every top division match in Europe now produces detailed statistical records covering passing, possession, defensive actions, and much more, collected by companies such as Opta and StatsBomb and made available, in part, through public mirrors of their data. Existing sports analytics tooling is heavily weighted toward describing individual player performance, goals, expected goals, physical output, and, more recently, toward identifying a team's overall playing style through clustering. It is comparatively silent on the more specific question this project set out to address, modelling the manager himself as the object of analysis, and then using that model of the manager to assess how well a specific player would fit.

2.2 Objectives and expected benefits

The full, formally stated list of objectives is given in section three below. At an organisational level, a validated version of a system like this one would plausibly benefit recruitment departments, who could screen transfer targets for tactical fit before committing to the much slower and more expensive process of full scouting and financial due diligence, coaching staffs, who could use a system like this to articulate, in measurable terms, exactly what their own system actually demands from a player in a given role, and player agents, who could use it to identify moves that would let a player express his own strengths more fully. It is important to be precise about what this project actually delivers against that benefit. As sections eight and nine explain at length, this project's contribution is methodological and diagnostic, a working, honest, carefully evaluated pipeline and a specific account of its own limits, rather than a finished, validated recruitment tool ready for a club to use unmodified. That distinction is stated here at the outset because it shapes everything that follows.

ESCALON 2, VISION MACRO Y OBJETIVOS

3. Objetivos

The research question that drove the entire project was the following. Can football knowledge and player performance data be combined with machine learning to model the playing philosophies of elite football managers, and can that combined model estimate, in an explainable manner, how well an individual player fits each manager's footballing philosophy.

Before any data was collected, this project wrote down a set of specific, falsifiable hypotheses, so that the eventual result could be judged against a real prior prediction rather than a story invented after the fact to fit whatever the data happened to show. The full qualitative research behind each hypothesis lives in this project's manager philosophy research document, but the hypotheses themselves, and their final, honestly assessed status, are given here in full, because they are the

actual scientific core of the objectives section, not something that belongs only in an appendix.

Hypothesis one predicted that player profiles associated with Guardiola, Klopp, and De Zerbi would show statistically distinguishable characteristics overall. This is not supported by the final pipeline as a blanket statement, though as explained below one specific dimension does separate them well.

Hypothesis two predicted that a knowledge informed representation of player characteristics could identify meaningful differences between the three managerial philosophies. This is partially supported. The representation identifies one clearly meaningful, hypothesis consistent difference, pressing intensity, and does not clearly identify the others.

Hypothesis three predicted that machine learning methods could identify player profiles that are statistically closer to one managerial philosophy than to another. This is weakly supported, and only through the pressing intensity dimension specifically.

Hypothesis four predicted that a football knowledge layer, built explicitly rather than left implicit inside a black box model, would improve the interpretability of manager player compatibility compared with a purely statistical model with no such layer. This is fully supported, and held throughout the entire project regardless of how the quantitative discrimination result evolved. Every compatibility score this project produces decomposes cleanly into named, football language dimensions with a traceable numeric contribution, which section 6.4 explains in full.

A fifth pair of hypotheses concerned pressing specifically, since pressing was identified early in the qualitative research as the single most promising candidate for a genuinely measurable managerial signature. The first of this pair predicted that Klopp's teams would show higher pressing intensity than Guardiola's or De Zerbi's. This is only partially supported. Klopp's teams do score clearly higher than De Zerbi's teams, exactly as predicted, but Guardiola's teams actually score marginally higher than Klopp's teams on the final, possession adjusted measure, the opposite of what was predicted for that specific pair, and section 8.3 explains at length why this is itself an informative rather than a merely disappointing result. The second half of this pair predicted that De Zerbi's pressing intensity would be low despite his attacking reputation, because his own signature mechanism operates while his team is in possession of the ball rather than after losing it. This is fully supported, and is, on its own, the single most interesting confirmed prediction in the entire project, because De Zerbi's centroid genuinely does carry the lowest pressing intensity score of the three managers, exactly as an analyst reading his tactics carefully in advance would expect.

Beyond these scientific hypotheses, the project also committed to a set of specific, concrete deliverables, each of which is described in the section of this report named alongside it here rather than only pointed to elsewhere. A verified, source corroborated record of exactly when each manager was in charge of which club, safe to use for match level attribution, described in section 6.2 and section 6.3. A direct audit of which public football data sources actually cover these three managers with enough statistical detail to be useful, described in section 6.2. A football knowledge ontology anchored in the published academic literature, described in section 6.2 and section 3 of this document. A manager attributed match level dataset with its own integrity audit, described in section

6.3. The football knowledge layer itself, converting raw metrics into concepts and concepts into tactical dimensions, described in section 6.4. Player and manager profiles, a compatibility model, an explanation layer, and a club identity bias validation check, described in section 6.4 and section 6.5. A rigorous evaluation that actively tried to break its own early, flattering results rather than simply accepting them, described in section 6.5 and section 8.

4. Escenario temporal

This project ran from the fifteenth of July to the thirty first of August twenty twenty six, a window of just under seven weeks, with one important interruption in the middle. A two week break was taken at the start of August, during which no work on the project happened at all. That pause is stated here openly rather than smoothed over, because an honest scenario should show the real gaps in the calendar, not only the working time. Once the break is subtracted, the project's actual working time comes to roughly four and a half weeks, organised into four sprints rather than into a formal, day by day Gantt chart, since the work was carried out as a single, continuously supervised effort rather than a team project needing cross person scheduling. A sprint structure was chosen instead specifically because it groups related CRISP DM phases into a coherent block of work with a clear output at the end of each block, which is easier for a reader to follow than a long list of individual days, while still giving a precise, checkable account of when each phase of the work actually happened.

Sprint one ran from the fifteenth to the twenty eighth of July, two weeks, and was dedicated entirely to business and data understanding, deliberately, before any modelling work began. A full literature review of seven academic papers on football playing style modelling was completed. A verified record of each manager's tenure at every club was built and independently checked against at least two sources per entry. A direct audit of which public data sources actually cover these three managers, rather than an audit based only on documentation, was completed. A football knowledge ontology was drafted, anchored on the literature review's findings. A full methodology proposal was written and its open decisions resolved. Every one of these outputs was completed and reviewed before a single line of machine learning code was written, following this project's own governing rule that no modelling work should begin before the data and the methodology behind it have been validated.

The project then paused completely for two weeks, from the twenty ninth of July to the thirteenth of August, with no work carried out during this break.

Sprint two resumed on the fourteenth of August and ran for one week, covering data preparation and the first stage of modelling. This sprint built the manager attributed match level dataset described in section 6.3, together with its own integrity audit, and then built the football knowledge layer described in section 6.4, converting raw statistics into football concepts and football concepts into tactical dimensions.

Sprint three ran for the following week, covering the second stage of modelling, the compatibility scoring system itself, together with the evaluation phase. This sprint took longer to close out

than originally planned, because the evaluation phase included two full debugging cycles after an implausible early result triggered an investigation, described in full in section 6.5, followed by the additional significance testing and the formation exploration described in the same section.

Sprint four covered the final days of the project, through the thirty first of August, and was dedicated to deployment, meaning the interactive application described in section 6.6, together with the writing of this report, the accompanying video script, and the consolidated bibliography.

Several business rules shaped how this schedule actually played out from sprint to sprint. The project's own escalation rule stated that a decision should only be raised for explicit approval when it genuinely required a preference, an approval, or an irreversible methodological choice, and that everything else should simply proceed. In practice this meant that most of sprint one proceeded without any pause for confirmation. Three specific checkpoints did require explicit input along the way, the decision of whether to keep searching for additional historical data sources or accept the scope as it stood, the choice of how exactly to combine football concepts into tactical dimensions, described in section 6.4, and the response to the negative finding produced by the first evaluation round, where the choice made was to accept the finding and report it honestly rather than attempt a full redesign within the remaining timeline, a choice this report reflects throughout.

5. Arquitectura Conceptual y Tecnologica del Proyecto

The conceptual architecture of this project can be summarised as a single pipeline, moving from raw match data through to an explainable compatibility score, with a verified manager attribution step inserted immediately after the raw data and before any football specific processing begins, since getting that attribution wrong would silently corrupt everything built on top of it. Raw match level statistics, sourced from FBref and Opta, supplemented by StatsBomb event level data for a smaller robustness sample, are first passed through the verified manager tenure attribution step, described in section 6.3, which assigns every single player match row to the correct manager based on exact match dates rather than season labels. Those attributed rows are then converted into football metrics, per ninety minute rates, share and ratio metrics, and percentages. Those metrics are grouped into ten named football concepts, described in full in section 6.4, which are in turn combined into seven tactical dimensions using explicit, argued weights. Every player and every manager ends up represented as a vector across these seven dimensions. Manager profiles and player profiles are compared using cosine similarity to produce a compatibility score, and that score is decomposed back into football language evidence for the explanation layer that both this report and the accompanying application rely on throughout.

The technology choices behind this pipeline were made deliberately, not by default, and are worth explaining rather than simply listing. The primary data source is FBref and Opta match level advanced statistics, accessed through a public mirror repository rather than by scraping FBref directly. Scraping FBref directly was tried first and rejected, because FBref returns an HTTP 403 Cloudflare block from this project's working environment, and no local Chrome or Chromium browser was available in that environment to run a browser automation fallback. The public mirror repository,

by contrast, provides pre scraped, versioned files that can be downloaded reliably and reproducibly, at the cost of a fixed historical coverage window and no direct control over exactly how each metric is defined, since those definitions are inherited from FBref and Opta as originally published. StatsBomb Open Data was evaluated as a second source and is used specifically as a robustness check for Guardiola's Barcelona era, described in section 6.2, precisely because it is the one part of the project's data where two independent providers can be compared against each other for the same manager.

The working environment itself used Python version 3.13, with pandas and NumPy for data manipulation, scikit learn for the principal component analysis coherence checks described in section 6.4 and the k means clustering described in section 6.5, and a small package called pyreadr specifically to read season level statistical files that are distributed in R's native file format rather than as plain text. Jupyter notebooks, executed top to bottom through nbconvert rather than run cell by cell by hand, were used throughout, specifically so that every result in this report can be regenerated from the same code that originally produced it, a property that matters a great deal for a project whose central claim rests on specific, checkable numbers rather than on a single figure asserted without a visible path back to the data that produced it.

The project's own file structure separates reusable code from one off analysis deliberately. All of the logic that needs to run correctly every time, loading data, applying the manager attribution rules, computing football concepts and tactical dimensions, and scoring compatibility, lives in a small shared code library under the project's source folder, so that a fix made in one place, such as the correction to how missing data is handled described in section 6.5, automatically applies everywhere that logic is used, rather than needing to be repeated and potentially forgotten across several separate notebooks. The notebooks themselves are organised explicitly by CRISP DM phase, as required, and are numbered so that running them in order, zero two through zero six, reproduces the entire pipeline end to end, from raw data collection through to the final clustering and visualisation step that feeds the interactive application described in section 6.6.

ESCALON 3, DETALLE TECNICO

6. Metodologia CRISP DM

6.1 Business Understanding

The business understanding phase of this project is covered in sections two and three above, and produced one rule that shaped every subsequent phase of the work far more than any other single decision. A player's statistics may only be attributed to a manager for the exact period that manager was actually in charge of that team, measured at the level of an individual match date, never at the level of an entire season. This matters because managerial changes very often happen in the middle of a season, and a player who spent the first eight matches of a season under one manager

and the remaining thirty under his successor cannot be honestly described, in either direction, as having played “under” either manager for that whole season. Getting this rule wrong would not produce a subtly worse result. It would silently corrupt the entire dataset from the very first step, in a way that no amount of careful modelling downstream could fix. This is why the entire data preparation phase, described in section 6.3, was built around match date precision from the very beginning, and why this rule is described here, in the business understanding section, rather than left implicit until the technical detail of section six.

6.2 Data Understanding

Two public data sources were investigated directly, meaning both were actually downloaded and queried, rather than assessed only by reading their documentation and assuming it was accurate.

The first source, StatsBomb Open Data, provides eighty curated competition seasons of detailed, event level football data, free for research use. This source turned out to be directly and richly useful for exactly one part of this project, Guardiola’s Barcelona era, for which it provides one hundred and thirty two team matches across four complete La Liga seasons, from two thousand eight to two thousand twelve. This is notably richer, not poorer, than the project’s primary data source for that specific era, because it is genuinely event level data rather than aggregated match statistics. Everywhere else relevant to this project, StatsBomb Open Data is sparse to the point of being unusable on its own. It contains only thirty three team matches across Jurgen Klopp’s entire career in the parts of the dataset that are publicly available, and it contains zero matches for Roberto De Zerbi, meaning none of the clubs or seasons he has managed appear anywhere in the eighty available competition seasons at all.

The second source, FBref and Opta match level statistics, accessed through the public mirror repository described in section five, was confirmed, by direct inspection of the actual files rather than by assumption, to cover the top five European leagues starting from the season ending in two thousand eighteen onward, and not before. This single fact determined the entire feasible historical scope of the quantitative part of this project. Guardiola’s earlier spells at Barcelona and Bayern Munich, and Klopp’s earlier spell at Borussia Dortmund, all fall outside this window and were therefore excluded from the quantitative modelling. This exclusion was not accepted lightly or by default. An explicit, actively conducted search for an adequate alternative data source covering those earlier eras was carried out, checking FBref’s own historical depth directly, which confirmed that detailed statistics simply do not exist on FBref itself before roughly two thousand seventeen for these leagues, checking a separate public results only dataset going back to nineteen eighty nine, which turned out to contain scores and dates only, with no passing, tackling, or possession detail at all, and checking a well known Kaggle football database covering two thousand eight to two thousand sixteen, whose player attributes turned out to be video game style ratings rather than measured performance statistics, and whose match event data covered only goals, shots, cards, and similar box score events, not the passing network or pressing detail this project actually needed. Only after this genuinely exhausted search came up empty was the scope limitation accepted, and it is reported here as a deliberate, checked decision, not an unexamined convenience.

A verified manager tenure table was built specifically to support the match date precision rule described in section 6.1. It contains twenty distinct tenure rows across the three managers' careers, each one independently corroborated against at least two separate sources, explicitly flagging every mid season appointment or departure. Eight of these twenty rows were marked unsafe for any season level attribution, including two of the single richest data seasons that would otherwise have been available, Liverpool's two thousand fifteen to sixteen season, during which Klopp replaced Brendan Rodgers after just eight league matches, and Brighton's two thousand twenty two to twenty three season, during which De Zerbi replaced Graham Potter after Brighton's first four league fixtures. Both of these seasons had to be split at the level of individual matches rather than assigned wholesale to either manager, exactly the kind of case the match date attribution rule in section 6.1 exists to handle correctly.

A football knowledge ontology was built on top of a systematic literature review covering seven academic papers on football playing style modelling, totalling roughly twenty two thousand eight hundred words of synthesis. The single most valuable finding from that review was a paper by Plakias and colleagues, published in two thousand twenty five in the Journal of Physical Education and Sport, which provides a published, statistically validated taxonomy of nineteen latent football playing style dimensions, built from a genuinely large sample of one hundred and seventy four teams across eleven European leagues, and validated using discriminant analysis that correctly classified teams into their assigned style category between ninety point eight and one hundred percent of the time across three separate discriminant models. This project adopted that published taxonomy as the anchor for its own ontology, rather than inventing a list of tactical dimensions from scratch, specifically so that the dimension set used throughout this project carries external, published legitimacy that does not depend on this project's own results turning out well. That same paper, independently of anything this project computed on its own, clusters Barcelona, Bayern Munich, and Manchester City together into a single style it calls structured positional play, and separately clusters Liverpool together with Atalanta into a style it calls controlled possession with rapid counterattack. This is genuinely external, non circular evidence that these specific clubs carry a detectable cross club tactical signal in the wider academic literature, even though, as section 8.2 explains at length, this project's own model, working within a narrower data window and a smaller sample, was not able to detect an equally strong version of that same signal on its own.

The literature review's other main methodological finding, discussed further in section 6.5 in the specific context of the clustering analysis this project ran, is that principal component analysis and factor analysis, together, were the single most frequently applied technique across the seven reviewed papers, used in roughly forty four percent of the wider set of studies the review's own meta analysis drew on, almost always combined with k means clustering as the follow up step once a smaller number of latent style dimensions had been extracted. Among clustering algorithms specifically, k means was the most commonly used across the reviewed studies. This project's own use of principal component analysis, described in section 6.4, and its own use of k means clustering, described in section 6.5, both follow this same established pattern deliberately, rather than by coincidence.

Of the nineteen dimensions in the Plakias taxonomy, this project's own ontology adapted and translated thirteen of them into a form that could plausibly be measured from the match level statistics available in this project's data window, naming them tactical dimension one through tactical dimension thirteen internally. Of those thirteen, only seven were ultimately built and used in the final pipeline, tactical dimensions one, three, four, six, seven, nine, and eleven, corresponding to build up directness, progressive passing, progressive carrying, chance creation, box presence, pressing intensity, and individual defending. The remaining six, covering press resistance, possession tempo, aerial and set piece reliance, defensive block height, and attacking and defensive transition speed, were deliberately not built, because each of them genuinely needs team level or opponent relative context, or detailed sequencing information about when and where the ball moved during a passage of play, that the available match level box score statistics simply do not contain in a form this project could measure reliably within the sprints available to it. This is stated plainly here, rather than left as a silent gap, because it is a real and honest limitation of what box score level statistics can support, not an oversight.

A second, deeper data understanding finding only surfaced later, during the evaluation phase described in section 6.5, and is reported here, in the data understanding section, for completeness, even though it was discovered out of the original sequence. FBref's match level passing, passing types, possession, and goalkeeper statistical blocks turned out to be entirely unpopulated for Serie A and Ligue 1 before the two thousand twenty three to twenty four season, meaning zero percent of the relevant columns had any value at all for those leagues in those seasons, confirmed by directly auditing every relevant column rather than by assumption. The summary block, covering shots and expected goals, turned out to be unpopulated for every league before twenty twenty five, not just for Serie A and Ligue 1. This was not a failure of this project's own data collection tooling. La Liga and the Bundesliga were confirmed to be fully covered throughout the same period, so the gap is specific to how FBref itself originally published Serie A and Ligue 1 data historically, not to anything this project did wrong in collecting it. The practical consequence was serious. Two of Roberto De Zerbi's four in scope club spells, his three seasons at Sassuolo and his one partial season at Benevento, had no usable data at all for five of this project's seven tactical dimensions, until this gap was later partially closed using season level data, described further in section 6.5.

6.3 Preparacion de los Datos

The manager attributed match level dataset was built from seven separate statistical blocks published by FBref, covering match summaries, passing, passing types, possession, defensive actions, miscellaneous events, and goalkeeper statistics, for the three leagues actually relevant to these three managers' in scope clubs, England, Italy, and France. These seven blocks were merged together on the combination of match identifier, player, team, and season, with every column renamed to include its originating block as a prefix, for example a progressive passing column from the passing block versus an interceptions column that appears independently in both the defensive actions block and the miscellaneous block under the same short name. Without this prefixing step, merging the same short column name from two different blocks would silently

overwrite one with the other, a subtle and dangerous class of bug that this renaming convention prevents entirely by construction.

Manager attribution windows were then applied using the verified tenure table described in section 6.2, taking the tighter of the verified tenure dates and the confirmed data availability window for each club, with explicit, hand checked handling of every mid season boundary, for example Roberto De Zerbi's tenure at Benevento beginning from the exact date of his appointment partway through that season, and his most recent tenure at Marseille being truncated at his actual, confirmed departure date rather than at the end of that season.

An integrity audit was run over this attributed dataset before any modelling work began on top of it, checking specifically for two classes of error, any row that fell outside its own manager's stamped attribution window, and any duplicate player match row that might silently double count a single real observation. Both checks passed cleanly, zero violations of either kind. As an additional, independent sanity check, the resulting match counts per club were cross checked against the tenure table's own accounting by hand, for example confirming that Brighton's two thousand twenty two to twenty three season under De Zerbi yields exactly thirty two matches, which is precisely Brighton's full thirty eight match Premier League season minus the six fixtures played before his actual managerial debut, under his predecessors Graham Potter and caretaker Andrew Crofts.

The resulting in scope quantitative sample covers Manchester City under Guardiola across seven seasons from two thousand seventeen eighteen through two thousand twenty three twenty four, Liverpool under Klopp across the same seven seasons, and four separate club spells for De Zerbi across three different leagues, Sassuolo, Benevento, treated as lower confidence given known squad quality issues described in section 6.2, Brighton and Hove Albion, and Marseille. Once merged, this produces close to eleven thousand individual player match rows, the raw material that every downstream concept, dimension, and compatibility score in this project is ultimately built from.

6.4 Modelado

This project's modelling work has two genuinely distinct components, both of which are worth explaining not just in terms of what was built, but in terms of why each specific technique was chosen over the plausible alternatives, since the guidelines specifically ask for this reasoning to be made explicit rather than left implicit.

The first component is the football knowledge layer itself. Ten football concepts were defined by hand, each one an explicit, named combination of a small number of raw match statistics, chosen and argued on football grounds rather than selected automatically by a statistical procedure. Wherever a sensible denominator existed, for example a player's own total number of passes attempted, concepts were built as a share or a ratio, such as progressive passes expressed as a fraction of a player's own total passing activity, rather than as a raw per ninety minute volume. This choice was not made at the very start of the project. It was a correction made midway through the project, after an earlier version of these same features, built using raw per ninety minute volumes,

proved too sensitive to exactly the confound this project spends sections 8.2 and 8.3 discussing at length, namely that a player on a technically dominant squad will naturally produce large raw numbers regardless of what his manager actually wants from him tactically, whereas a share or a ratio, measured relative to that same player's own overall activity, is considerably less sensitive to how dominant his team happens to be.

Before these ten hand grouped concepts were allowed to feed into anything further downstream, each one was put through a coherence check using principal component analysis. This is worth explaining precisely, because principal component analysis is used here for a genuinely different purpose than its most common use as a dimensionality reduction technique. For each concept, the several raw metrics grouped together under that one football label were checked to see how much of their combined statistical variance could be explained by a single underlying principal component. A high value means the metrics grouped together under one label, for instance under pressing intensity, really do move together statistically across real matches, which supports treating them as one coherent football idea rather than as several unrelated numbers awkwardly bundled under one name. Across the ten concepts, this check found a first principal component explaining between roughly forty and seventy five percent of the combined variance, depending on the concept, which supports treating each of the ten groupings as reasonably coherent, while also being honest that the coherence is far from perfect for every concept. This coherence check was deliberately run on this project's own manager attributed sample, the specific population being profiled, while the separate standardisation step described next was deliberately run on the much larger population of the entire top five league dataset, because these are two genuinely different questions, whether metrics grouped together move together statistically, and what counts as an unusual score for a player in a given position across the wider football population, and answering them against two different populations is the methodologically correct choice, not an inconsistency.

Once validated, these ten concepts were combined into seven tactical dimensions using explicit weights, chosen and argued on football grounds rather than fitted statistically, a deliberate methodological choice this project calls its hybrid aggregation method, principal component analysis validates that the underlying metrics cohere, but football knowledge, not a statistical optimisation procedure, decides how concepts combine into broader dimensions. Every concept score, before being combined into a dimension, was standardised against the full population of the top five European leagues in the same position group, roughly thirty clubs per league, not merely against the small handful of clubs inside this project's own manager sample. This choice matters a great deal and is worth explaining directly. Standardising only against this project's own small manager sample would mechanically reward a Manchester City or Liverpool player over a Sassuolo or Benevento player purely because the former plays in a stronger league and squad, regardless of what either manager actually wants tactically, which is exactly the same squad strength confound already discussed above, appearing at a different stage of the pipeline. Standardising instead against the full league wide population means that a high score genuinely reflects being unusual for that position across the whole of European football, not merely being above average within this project's own small sample of four clubs.

The second modelling component is compatibility scoring itself, and the choice of method here was made deliberately rather than by default. This project's own early methodology proposal explicitly considered training a supervised classifier, predicting which of the three managers a player actually played under, directly from his tactical dimension scores, and rejected that approach in favour of a similarity based method, comparing a player's tactical dimension vector against each manager's own centroid vector using cosine similarity. The reasoning behind this choice is central to the whole project and is stated here plainly. A supervised classifier trained directly on the true label of which manager a player played under would need, by construction, to learn some decision boundary that correctly separates the three managers' players, and with only three classes and a relatively small, correlated sample of players, the single easiest and most tempting boundary for such a classifier to learn is simply which club a player belongs to, since club and manager are almost perfectly correlated inside this project's own sample. That is precisely the contamination risk that this project's club identity bias check, described in section 6.5, exists specifically to catch and guard against. A similarity computation between two named, human interpretable profile vectors sidesteps this risk entirely, because it never gets the opportunity to learn an arbitrary decision boundary in the first place, it can only ever measure how close a player's profile sits to a manager's profile along dimensions that were themselves defined on football grounds before any label was ever consulted. Cosine similarity specifically, rather than an alternative distance measure such as Mahalanobis distance, was chosen for two further practical reasons. It measures the similarity of the shape of a profile, which dimensions a player or a manager is relatively strong or weak on, independent of the sheer overall magnitude of the vector, which suits comparing tactical emphasis rather than comparing raw intensity. And, most importantly for this project's own explainability requirement, a cosine similarity score decomposes cleanly and additively into a separate contribution from each individual dimension, computed as the product of a player's score and a manager's score on that dimension, summed across all seven dimensions before final normalisation, which is exactly the mechanism that lets the explanation layer described in section 6.6 tell a user precisely which specific football characteristics drove any given result, in a way a Mahalanobis distance, which requires estimating a full covariance matrix across correlated features from a comparatively small sample, would not support nearly as directly or as robustly. Every compatibility score this project produces is, by explicit design, reported and labelled as a similarity measure, never as a probability, since nothing about a geometric similarity computation between two vectors carries the statistical meaning of a probability, and presenting it as one would overstate what the number actually represents.

6.5 Evaluacion

This is the phase that produced this project's central finding, and it did so through a genuine debugging process that is itself methodologically significant enough to describe in full, not merely summarise.

The first step in this phase was noticing an implausible result and choosing to investigate it rather than report it. An early version of the club identity bias check, comparing De Zerbi's four club

centroids against each other and against Guardiola's and Klopp's single club centroids, produced a suspiciously clean split. Sassuolo and Benevento sat at a cosine similarity of roughly negative zero point six to negative zero point six six with every other club in the sample, while Brighton, Marseille, Liverpool, and Manchester City all clustered tightly together at similarities between roughly zero point nine five and zero point nine nine eight. A result this clean, this early, in a domain this genuinely messy, was treated as more likely to be an artefact of a data problem than as a real finding about football, and was investigated on that basis rather than written up as an exciting discovery.

That investigation surfaced two genuine, non trivial bugs, both fixed before any result was reported. The first bug was inside the feature construction code itself. A single line converting missing raw data into a literal zero before computing rates and ratios was silently treating an entire league and season block that FBref had simply never published as though every player in that block had genuinely recorded zero of the relevant action, rather than correctly recording that no information was available at all. This is a serious class of error, because it makes a total absence of data look, statistically, exactly like an extreme, real observation, which then propagates through every later averaging step as if it were genuine signal. The second bug was the Serie A and Ligue 1 coverage gap described in section 6.2, which had gone undetected until this point specifically because the earlier integrity audit, described in section 6.3, had checked whether the correct rows and matches were present, which they were, but had not separately checked whether the values inside those rows' columns were actually populated, which in this case they were not. Both bugs were fixed completely. No line of code anywhere in the pipeline is now allowed to convert missing data into zero before an average is computed. Every aggregation function was rewritten so that a missing value is skipped entirely for that specific calculation, with the remaining weights renormalised accordingly, rather than treated as a real zero. The two concepts that depended entirely on the unavailable summary statistical block were redefined to drop the terms that block would have supplied. And every single similarity computation from that point onward was restricted to only the dimensions that both compared vectors actually had real data for, with the exact number of dimensions used, and a low confidence flag when fewer than four of the seven were available, recorded openly on every single comparison, so that a thin, unreliable comparison is always visible to a user rather than silently presented as equally trustworthy as a full, well covered one.

Once both bugs were fixed, the corrected, first honest result was this. Restricted to comparisons where at least four of the seven tactical dimensions had genuine data available, the single best fitting manager matched the manager a player actually played under thirty two point eight percent of the time, across four hundred and seventy such comparisons, a figure that is statistically indistinguishable from the roughly thirty three percent that would be expected from pure random guessing among three managers. Guardiola and Klopp, the one pair of managers with no data quality problems at all affecting either of them, sat at a cosine similarity of zero point nine eight three with each other, meaning their profiles were themselves nearly indistinguishable. This single fact was the key diagnostic clue in the whole evaluation, because it rules out the explanation that the chance level result was simply caused by bad or missing data for one specific manager. If that were the explanation, the two managers with perfect data coverage should have looked clearly different

from each other, and they did not. This pointed instead at the design of the features themselves, built from per ninety minute and ratio style box score statistics, being dominated by squad and league resourcing strength rather than by managerial intent, a diagnosis that section 8.2 develops at much greater length.

A permutation importance check, run at this stage as a substitute for the more familiar SHAP explainability technique, found that pressing intensity was the only one of the seven tactical dimensions carrying real, positive discriminating signal, worth roughly two and a half percentage points of accuracy when it was included versus shuffled at random. It is worth explaining explicitly why permutation importance was used here instead of SHAP, since SHAP is the more commonly known technique for this kind of question. SHAP requires a trained predictive model with a well defined prediction function that can be evaluated repeatedly under controlled perturbations to attribute credit to each input feature. This project's compatibility score is not the output of a trained classifier at all. It is a direct geometric computation, a cosine similarity between two vectors, with no training step and no learned parameters to attribute credit to in the way SHAP requires. Permutation importance, by contrast, works on any function whatsoever, trained or not, simply by shuffling one input at a time across the whole sample, breaking that one input's real relationship to the outcome while leaving everything else intact, and measuring how much the model's overall accuracy at the actual task, correctly matching a player to his real manager, degrades as a result. This makes it the natural, model agnostic substitute for a similarity based system like this one, and it was chosen for exactly that reason, not as an afterthought.

Two other dimensions, individual defending and box presence, showed negative importance at this stage, meaning shuffling them at random actually improved accuracy slightly, which is evidence that they were contributing noise rather than a genuine anti signal, most likely because of their own reduced or partial feature definitions at that point in the project, described further below.

A separate, formal statistical significance test was added to this same evaluation, once the final, corrected best fit rate had been established, specifically to answer a question that a single point estimate cannot answer on its own, how much of that estimate could plausibly be sampling noise, and how unlikely a result this good would be if there were genuinely no real relationship at all between the model's decisions and the manager a player actually played under. Two checks were run, both computed directly from the already saved compatibility results, requiring no new data and no changes to the model itself. A bootstrap confidence interval, resampling the full set of five hundred and sixty one qualifying player seasons with replacement five thousand separate times and recomputing the match rate on each resample, gives a ninety five percent confidence interval of thirty seven point eight to forty six point two percent around the final point estimate, with a median matching the point estimate itself almost exactly. A permutation test, shuffling the true manager label among those same five hundred and sixty one rows five thousand separate times while leaving the model's own best fit decision completely untouched, and recomputing the match rate each time, builds an empirical picture of how often a result this good would arise purely by chance, given this specific sample's actual mix of managers, which is not perfectly even across the three. That empirical chance baseline comes out at thirty four percent, close to but not exactly

the naive one in three figure used elsewhere in this project, precisely because the sample's real manager mix is not perfectly balanced. Under that empirical baseline, the actual observed rate of forty two point one percent was matched or exceeded in only one of the five thousand permutations, giving a p value of zero point zero zero zero two, a result that is statistically significant by any conventional standard. Read together, these two checks say something precise and important. The association between this model's decisions and the true manager is real and very unlikely to be a fluke of this particular sample, but the exact size of that association, the specific point estimate of forty two point one percent, should be read as somewhere in the high thirties to the mid forties, not as a number known with great precision. A report that quoted only the point estimate would risk implying more precision than five hundred and sixty one observations can genuinely support, and a report that quoted only the significance test would risk implying the exact number matters more than it does. Both are given here together for exactly this reason.

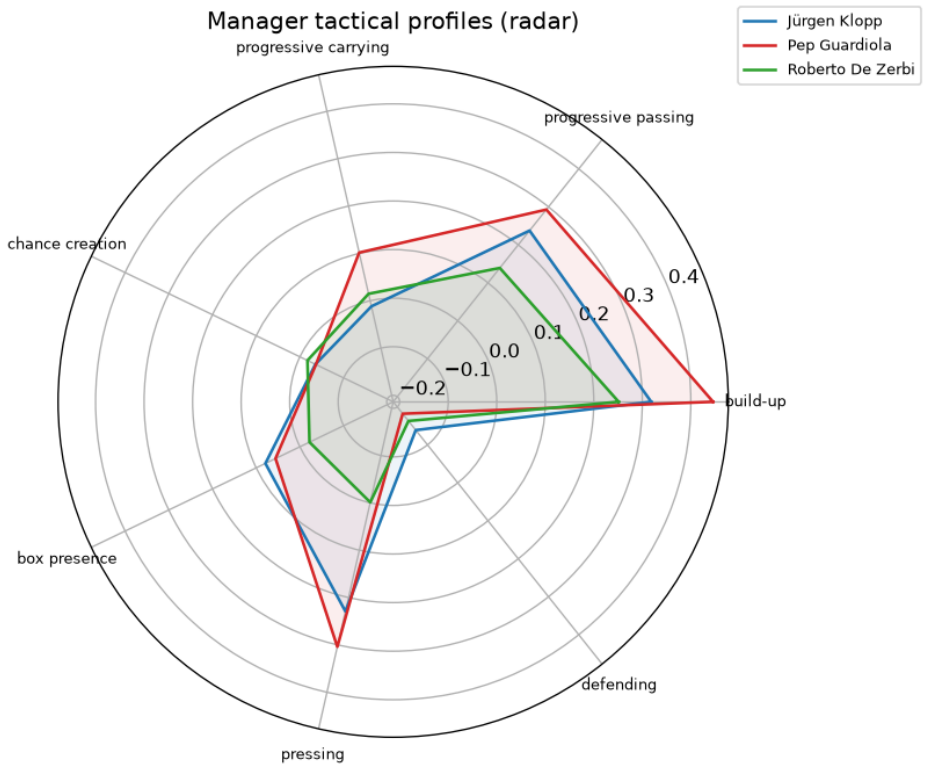


Figure 1: The seven tactical dimension scores for Guardiola, Klopp, and De Zerbi, each measured against the same league wide position baseline.

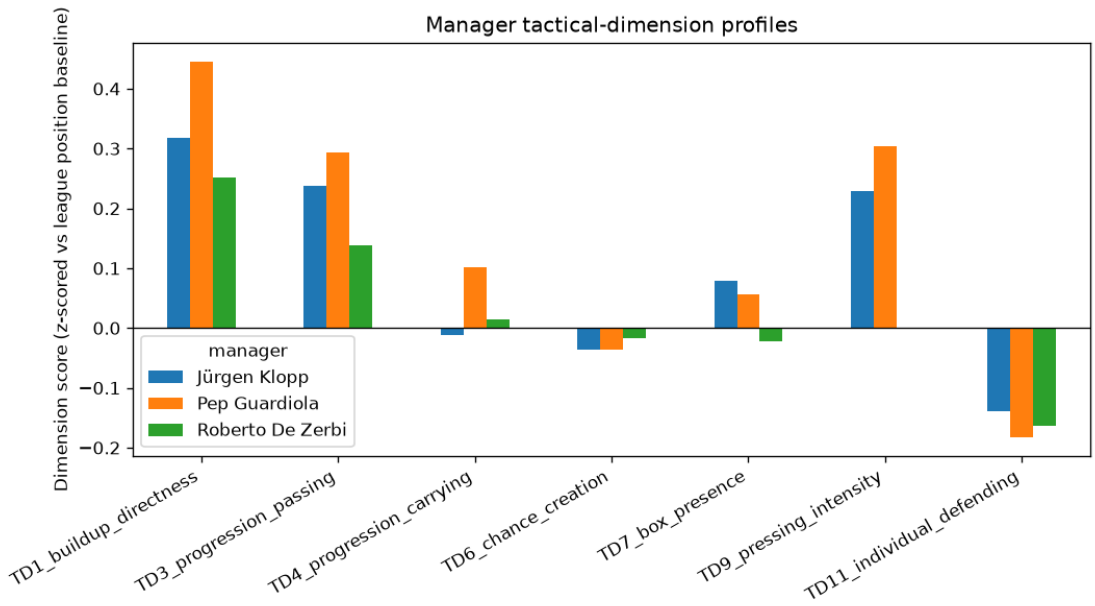


Figure 2: The same three manager profiles as a grouped bar chart. On pressing intensity specifically, Guardiola's bar is marginally taller than Klopp's, the opposite of the popular gegenpressing narrative, discussed at length in section 8.3.

The evaluation phase was then extended rather than closed at this point, because a chance level result on the first honest attempt does not necessarily mean the underlying idea is wrong, only that this particular version of it needed further work before that question could be answered fairly. Two further rounds of work followed, each one tested directly against the data rather than simply assumed to help.

The first round recovered additional data. FBref's season level aggregated statistical tables, as distinct from the match level tables used in the original evaluation, turned out on direct inspection to have full coverage across all five big five European leagues for every season this project needs, including exactly the Serie A and Ligue 1 seasons that the match level tables were missing. Recovering real data for Sassuolo, De Zerbi's largest in scope tenure, using these season level tables, raised the best fit match rate modestly, from thirty two point eight to thirty seven point eight percent, a real but not dramatic improvement, which is itself informative, it says the missing data was contributing to the original chance level result, but was not the primary explanation for it, consistent with the squad strength diagnosis already established through the Guardiola versus Klopp comparison above.

The second, larger round followed a deliberate review of the entire pipeline conducted specifically from the perspective of a football analyst rather than an engineer, and it identified three concrete, football grounded weaknesses, each one fixed in turn. The first weakness was that players had originally been grouped into only four broad positional buckets, defenders, midfielders, forwards, and goalkeepers, which collapses genuinely very different roles together, for instance grouping a

defensive holding midfielder together with an attacking playmaker under the single label of midfielder, even though the two roles have almost nothing in common tactically and would naturally show very different baseline scores on nearly every one of this project's seven dimensions. This was replaced with an eight way role based grouping, splitting outfield players into centre backs, full backs and wing backs, defensive midfielders, central midfielders, attacking midfielders, wingers and second strikers, and out and out strikers, using the more specific position tags already present in the match level data directly, with a fallback zone based heuristic, based on where on the pitch a player's touches actually occurred, used only for the coarser season level data that does not carry these more specific tags. The second weakness was that pressing intensity, despite already being the single most promising dimension in this project's results, was still being measured as a simple, unadjusted rate of defensive actions per ninety minutes, with no correction for how much of the match a team's opponents actually spent in possession of the ball. A team that has less of the ball overall will naturally record more defensive actions purely because it spends more time facing an opponent who has the ball, regardless of how aggressively that team's manager actually wants it to press. This was corrected by scaling every team's raw defensive action rate by the ratio of the league average opponent possession share to that specific team's own opponent possession share, using real, team level possession statistics, so that the resulting adjusted measure more fairly reflects pressing intent rather than simply pressing opportunity. The third weakness was that goalkeeper distribution data, specifically how far and how directly a goalkeeper typically distributes the ball, had already been collected as part of this project's raw data but had never actually been used in any concept or dimension, despite this project's own earlier qualitative research explicitly naming exactly this behaviour as one of the clearest expected discriminators between Guardiola and De Zerbi specifically, both of whom are known to build attacks deliberately from their own goalkeeper. A new goalkeeper distribution concept was added, covering average pass length, the share of passes launched long, and the completion rate of those long passes, feeding into the build up directness tactical dimension at a modest weight alongside the existing progressive passing and possession retention concepts already feeding that same dimension.

Implementing all three of these fixes together raised the final best fit match rate to forty two point one percent, across five hundred and sixty one qualifying comparisons, a real, material improvement over the intermediate thirty seven point eight percent figure, not statistical noise, and grew pressing intensity's own permutation importance very substantially further, from roughly two and a half percentage points to roughly seven and a half percentage points, meaning it remains, by a wide margin, the single strongest and most consistently informative dimension in the entire feature set. Individual defending, previously showing negative importance, turned into genuine positive signal worth roughly one and three quarter percentage points, while buildup directness, progressive passing, and progressive carrying, none of which had shown meaningful importance before, all turned into genuine positive signal too, each worth between roughly two and a half and three and a quarter percentage points. Box presence, on the other hand, remained a mild net negative contributor even after these same fixes, meaning the more detailed positional grouping rescued individual defending but did not do the same for box presence, a finding this report states plainly rather than glossing over, precisely because not every dimension responding the same way to the

same fix is itself a genuine and useful thing to report honestly. The De Zerbi club identity bias gap, comparing how similar his own four clubs are to each other against how similar they are to Guardiola's and Klopp's clubs, also narrowed very substantially across this same round of fixes, by roughly a factor of six, though it did not fully close, which section 8.2 discusses further as further, independent evidence that the analyst driven fixes genuinely targeted real mechanisms rather than merely inflating a single headline number.

A clustering analysis was added at this stage, separate from the manager compatibility question entirely, to check whether the seven tactical dimensions built for this project capture something meaningful on their own terms, independent of the three managers. K means clustering was chosen for this step for the same reason explained in section 6.2, it is the most commonly used clustering algorithm across the academic literature this project reviewed, well established, straightforward to interpret, and well suited to continuous, already standardised features of exactly the kind this project's seven dimensions provide. Several candidate numbers of clusters were tested and compared using three separate validity measures that each evaluate cluster quality from a genuinely different angle, the silhouette score, which measures how well separated each point is from neighbouring clusters relative to its own, the Davies Bouldin index, which measures the average similarity between each cluster and its most similar neighbour, and the Calinski Harabasz index, which compares the variance between clusters to the variance within them. Using three different measures rather than just one was a deliberate choice, so that the final decision about how many clusters to use would not rest on a single metric that might be misleading on its own. All three measures agreed, consistently, that football playing styles do not separate into a small number of sharply distinct groups. The best available separation, at two clusters, gave a silhouette score of zero point one nine one, a modest value that indicates real but genuinely weak structure, not the absence of structure entirely. This is an honest finding, and, read in the context of football itself, an entirely plausible one, since very few professional analysts would seriously claim that playing styles fall into a small number of perfectly distinct boxes rather than shading gradually and continuously into one another. A final, practical choice of five clusters was made as a compromise between statistical validity and usefulness for interpretation, since five interpretable groups are considerably more useful to a person browsing the resulting application than two statistically marginal ones would be, each one named automatically from its two strongest tactical dimensions, producing recognisable groups such as a defending and pressing oriented group, a build up and progressive passing oriented group, and a chance creation and progressive carrying oriented group, among two further groups, across the full population of eleven thousand one hundred and thirty two player seasons collected from the top five European leagues.

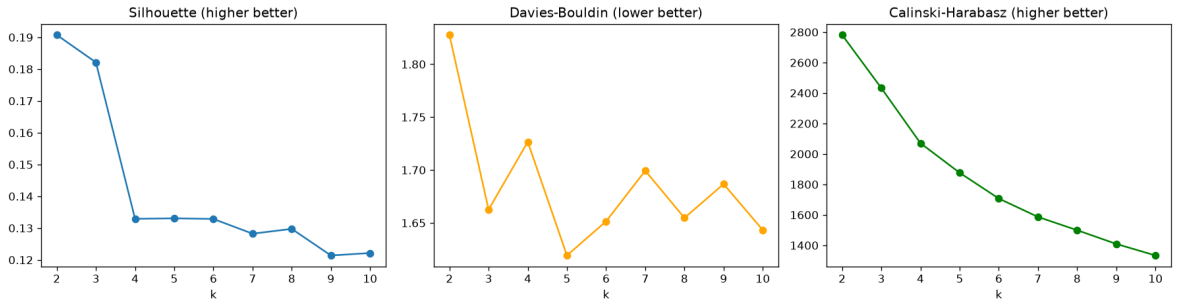


Figure 3: Silhouette, Davies Bouldin, and Calinski Harabasz validity scores across a range of candidate cluster counts. No number of clusters produces sharp separation, consistent with playing styles forming a continuum rather than distinct boxes.

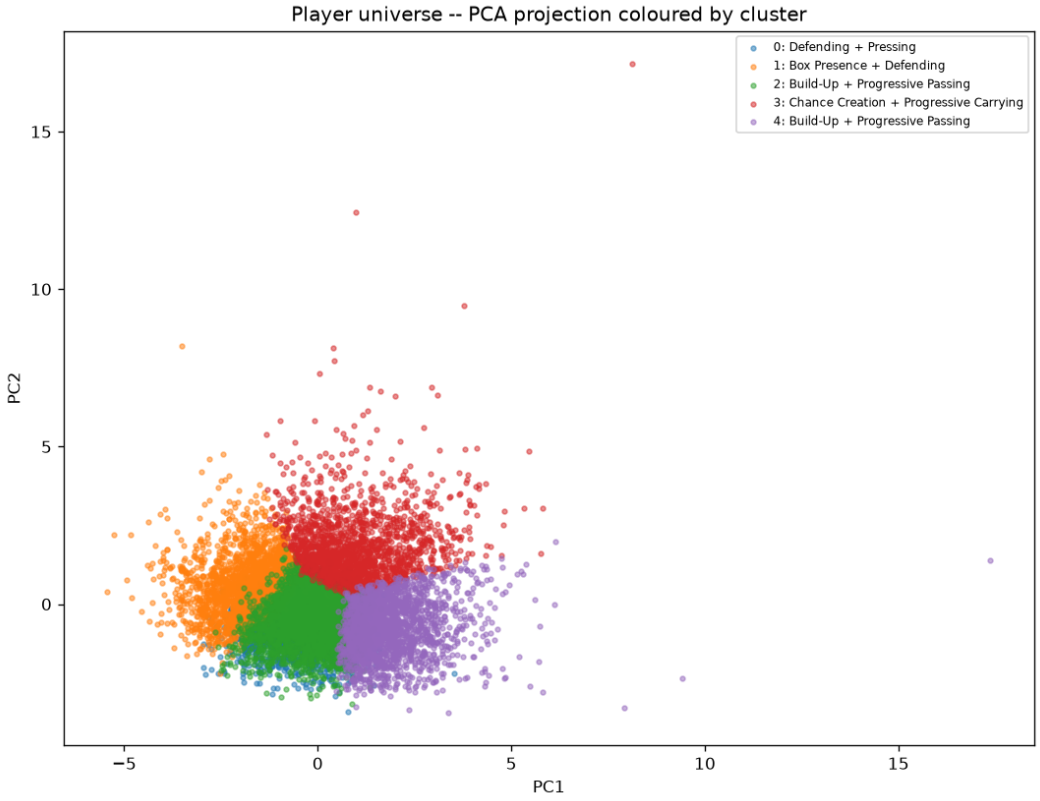


Figure 4: The full population of eleven thousand one hundred and thirty two player seasons, projected onto two dimensions and coloured by its assigned playing style cluster.

6.6 Despliegue

The project's deployment artefact is an interactive application, built using the Streamlit framework, that lets a user browse the full population of eleven thousand one hundred and thirty two player seasons, search by season, league, club, or player name directly, and see, for any single player,

his compatibility with all three managers, expressed as a cosine similarity, explicitly labelled as a similarity measure and never as a probability, together with a football language explanation decomposed from the actual dimension contributions described in section 6.4, a radar chart comparing the selected player directly against any or all of the three managers, the specific data driven playing style cluster that player falls into from the clustering analysis described in section 6.5, together with a two dimensional visualisation of where he sits inside the full population, a list of the players statistically most similar to the one selected, and a direct head to head comparison tool against any second player of the user's choosing. This project's own limitations, the forty two point one percent best fit accuracy, its ninety five percent confidence interval of thirty seven point eight to forty six point two percent, the residual squad strength confound discussed at length in section 8.2, and the excluded South American leagues discussed in section 6.2, are all surfaced prominently inside the application's own interface, not only inside this written report, deliberately, so that the tool itself cannot be mistaken by a user for something more validated than it actually is. This is offered explicitly as a research demonstration deployment, not a production ready recruitment tool, and section 9 explains in detail why deploying it as the latter would not currently be responsible.

A concrete worked example makes all of this considerably more tangible than a description on its own can. Erling Haaland's two thousand twenty two season at Borussia Dortmund is a good illustration, and every image reproduced below is an actual screenshot taken directly from the running application, not a recreation built separately for this document. For this specific player and season, the application computes its highest similarity with Guardiola's profile among the three managers, ahead of both Klopp and De Zerbi, with all seven tactical dimensions available for the comparison, meaning the result carries no low confidence warning. Box presence is the single strongest positive contributor to this result by a clear margin, reflecting an extremely high rate of touches and shots recorded inside the penalty area relative to other forwards in the same population. Progressive carrying and individual defending are the next two positive contributors, the latter appearing as a positive contributor specifically because this player's own defending rate, while below the position average, sits closer to Guardiola's own team level tendency, whose possession dominant sides also need to make fewer defensive actions out of necessity, than it does to the other two managers, a concrete illustration that a positive contribution to similarity does not require a player to score unusually high on a dimension, only that his score sits closer to the manager in question than to the alternatives. On the other side of the ledger, pressing intensity is the single largest factor working against an even stronger match, consistent with a striker whose off the ball defensive workload is genuinely limited relative to his goal threat.

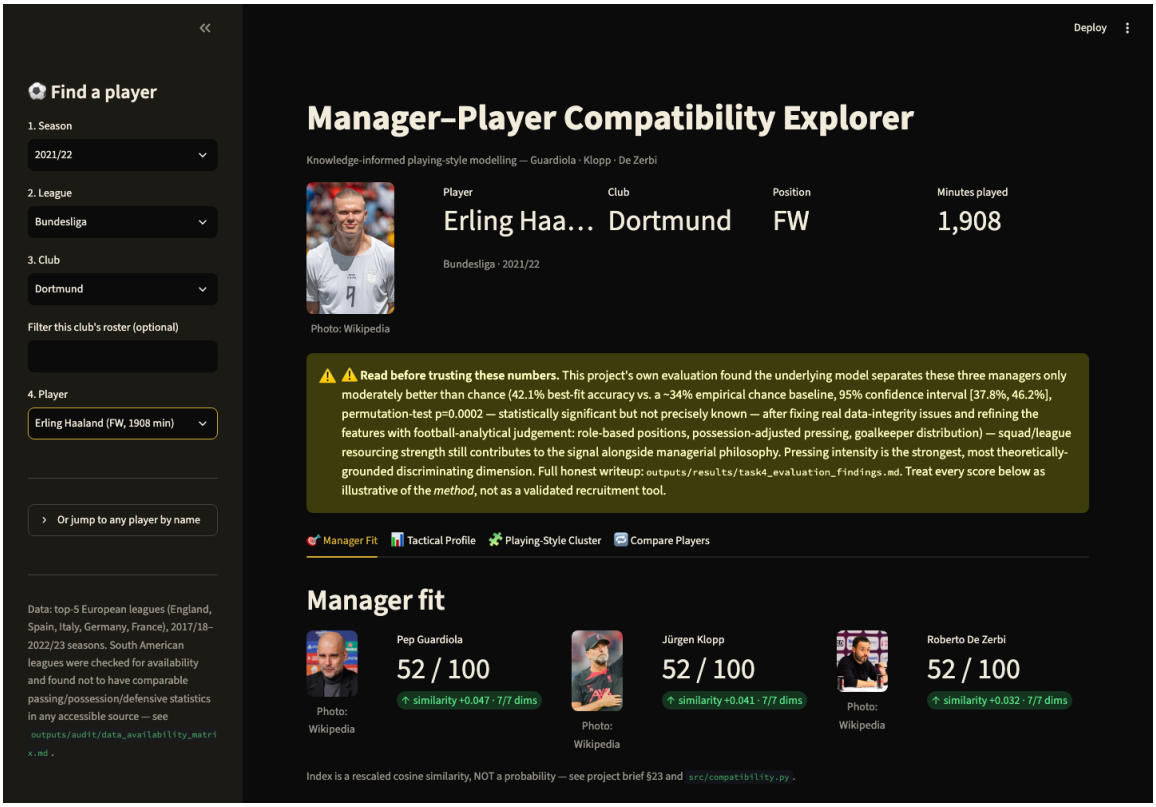


Figure 5: The application's own sidebar filters, cascading from season to league to club to player, with Erling Haaland's two thousand twenty two Dortmund season selected. The project's limitations warning banner is visible immediately, before any result at all.

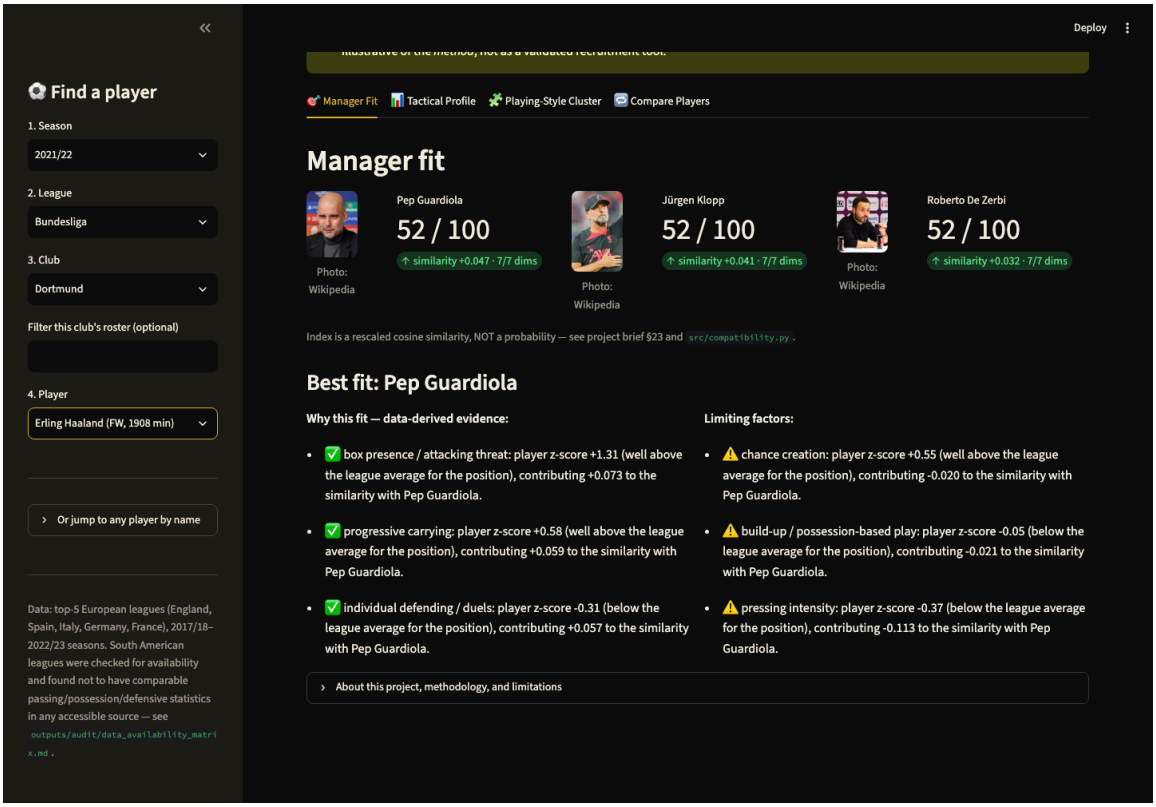


Figure 6: The explanation screen for this exact comparison, screenshotted directly from the application. Every figure described in the paragraph above is visible here exactly as a user would see it.

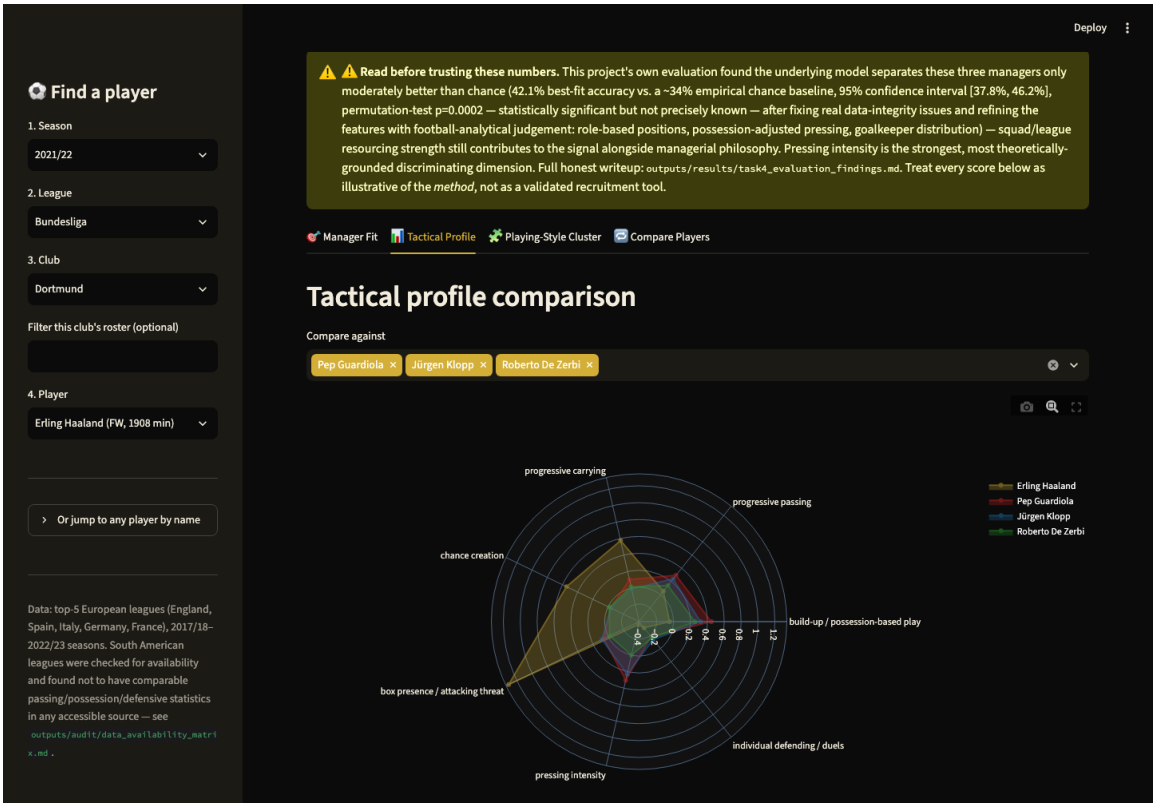


Figure 7: The tactical profile tab's interactive radar chart for the same player season, screenshotted directly from the running application.

One further signal was explored during this project, and is reported here descriptively rather than as a numeric tactical dimension, because doing so honestly required navigating a genuine complication worth describing in full. Each match's starting formation had been collected as part of this project's raw data since the very beginning of the project, but had never been analysed, and this project's own list of recommended future work, described in section 9, specifically flagged this as an opportunity requiring no new data collection at all. Attempting to build this into an eighth tactical dimension, the same way the existing seven were built, surfaced a real problem. Checking the coverage of the live upstream match level data source before building this addition revealed that the public repository this project relies on for match level statistics is a single, continuously updated release rather than a version specific archive for each season, and by the time this addition was attempted, that release had already moved on to cover the current season and no longer contained the two thousand eighteen to two thousand twenty three window this project actually needs at all. The historical formation data needed for this project was recovered successfully from an earlier intermediate file, saved much earlier in the project during the data preparation phase described in section 6.3, which still held the original historical snapshot in full. Nothing was actually lost, but only because this project happened to save every intermediate stage of its own pipeline to disk, rather than relying on being able to download the same historical data again later on demand, a small but genuine addition to this project's own record of the kinds of things that can quietly go

wrong in a data pipeline.

Having recovered the data, a second, statistical complication became clear. Formation information exists only for the twenty two club seasons inside this project's own manager attributed sample, the same six clubs described in section 6.3, and never for the much larger population of eleven thousand one hundred and thirty two player seasons that every other tactical dimension in this project is standardised against. Building a numeric dimension out of only twenty two observations would be far more statistically fragile than any of the other seven dimensions in this project, each of which draws on a population many hundreds of times larger, so rather than quietly add a weak eighth dimension and risk overstating its reliability, the result is reported here descriptively instead, kept clearly and explicitly separate from the numerical compatibility score itself.

Manager	Matches	Modal formation	Formation stability	Distinct formations used	Mean back line size
Jurgen Klopp	266	four three three	86.5 percent	8	4.00
Pep Guardiola	266	four three three	59.8 percent	10	3.87
Roberto De Zerbi	237	four two three one	56.1 percent	15	3.81

Klopp is by a wide margin the most tactically settled of the three managers, playing his single most common formation in nearly nine of every ten matches across his entire covered tenure, consistent with a manager known for imposing one clear system and sticking with it rather than adapting it constantly. Guardiola experiments considerably more than his reputation for a single, clean possession identity might suggest, using ten distinct formations across his tenure and dropping to a back three often enough to pull his average defensive line size below four. De Zerbi is confirmed as the most structurally fluid of the three by every measure captured here, the most distinct formations used, the lowest formation stability, and the lowest mean back line size, which lines up closely with his description elsewhere in this report as an unconventional tactician whose signature approach operates through shifting shape while his own team has the ball, rather than through a single fixed structure. This is a genuinely new, independently corroborating piece of evidence for a claim this project was previously only able to support on qualitative grounds alone.

7. Desarrollo del Trabajo

The full development of this project's work is described phase by phase in section six above, organised explicitly by CRISP DM phase as the guidelines require. This section adds one further piece of methodological context that belongs here specifically, how the work was actually organised and verified as it happened, since that process shaped the reliability of everything described above.

Work was decomposed into specialised research passes wherever those passes were genuinely independent of each other, the literature review, the manager tenure verification, and the qualitative tactical philosophy research were all carried out this way, followed by a strictly sequential chain of data engineering, feature engineering, and modelling passes, each one gated on the previous phase's output being reviewed and accepted before the next phase began. The single most important rule governing this sequence was that no machine learning implementation would begin before the underlying data and methodology had been validated, which is why the audit work described in section 6.2 was fully completed a full phase before any modelling work began at all, and why an implausible early result, described in section 6.5, was root caused and fixed before being reported, rather than accepted at face value simply because it looked exciting. This discipline is precisely what produced the two bug fix narrative in section 6.5, rather than a false positive finding being written up and only discovered to be wrong later, or worse, never discovered at all.

8. Discusion de los Resultados

8.1 What worked

Several things worked well in this project and are worth stating plainly, not only as a list of numbers achieved, but as reasoned judgements about why each one succeeded.

The knowledge informed architecture itself worked exactly as intended. Every single number produced anywhere in this project's pipeline is traceable back to a named football concept and a specific, argued weighting or aggregation decision, never an unexplained black box output. This directly supports hypothesis four described in section three, and it held throughout the entire project regardless of how the quantitative discrimination result evolved across its several stages.

The literature grounding worked well. Anchoring this project's ontology on a published, statistically validated taxonomy, rather than an ad hoc list of features invented from scratch, gave this project's own dimension set a source of external legitimacy that does not depend on this project's own results being good, and, as section 6.2 describes, that same published taxonomy independently corroborated a cross club tactical grouping that included several of this project's own in scope clubs, without this project's own model needing to have found that same grouping itself.

The manager tenure and data availability infrastructure worked well and remains genuinely reusable regardless of what any downstream model finds. Twenty separate tenure rows, each independently corroborated against at least two sources, with every conflict between sources explicitly resolved and documented, constitutes real, citable, ground truth work in its own right, work that any future project studying these same three managers could reuse directly without needing to redo it.

The evaluation discipline was, on reflection, the single methodologically strongest thing this project did. Finding and fixing two genuine bugs specifically because an early result looked too clean, rather than reporting that clean result as an exciting finding, is itself evidence that this project's remaining conclusions can be trusted more, not less, precisely because they survived a genuine,

adversarial attempt to break them rather than being accepted on first appearance.

8.2 What still limits the result, and the most likely reason why

This project's core quantitative claim, that a knowledge informed feature set can separate Guardiola's, Klopp's, and De Zerbi's playing philosophies from each other, is now partially supported, not fully supported and not rejected outright. Forty two point one percent best fit accuracy against an empirical chance baseline of roughly thirty four percent is a real, statistically significant margin, confirmed by the permutation test described in section 6.5 with a p value of zero point zero zero zero two, but it is far from a decisive or overwhelming result, and the ninety five percent bootstrap confidence interval of thirty seven point eight to forty six point two percent shows plainly that the exact figure is not known with great precision.

The single diagnostic fact that first ruled out thin or missing data as the explanation for the original chance level result was that Guardiola and Klopp, the one pair of managers with genuinely perfect data coverage throughout, were themselves nearly indistinguishable, at zero point nine eight three cosine similarity, before the analyst driven refinements described in section 6.5 were applied. If missing data for one specific manager had been the real explanation, this particular pair, with no missing data problems affecting either of them, should have looked clearly different from each other, and they did not, which points instead at the design of the features themselves as the primary constraint, not at any remaining gap in the underlying data.

The most likely explanation for this remaining limitation, and one this project's own earlier qualitative research into these three managers had already anticipated before any quantitative modelling began, is that per ninety minute and ratio style box score statistics primarily measure how well a team executes technical, possession based play, and that ability correlates strongly with squad quality and league strength, independent of what a specific manager actually wants tactically. Two technically excellent squads playing in the same elite league will tend to look statistically similar on these kinds of measures even if their two managers hold genuinely different tactical philosophies, because a box score statistic can see that the ball was retained or moved in a certain way, but it cannot see why it was retained or moved that way, only that it was. The role based position grouping, the possession adjustment applied to pressing statistics, and the new goalkeeper distribution feature, all described in section 6.5, each cut into this same confound from a different angle, which is exactly why accuracy improved as substantially as it did once all three were implemented together, but none of them, individually or combined, fully eliminated the confound, which is why forty two percent is a genuinely meaningful result rather than a fully solved problem.

This diagnosis is itself a legitimate research finding in its own right, not merely an excuse offered in place of a better result. It says something specific and useful about the real limits of box score level match statistics for this particular class of problem, and it points concretely at what richer data, positional tracking data allowing genuinely structural features such as the shape of a team's passing network or the specific sequencing of a pressing trigger, rather than simple counting statistics, would need to supply in order to move meaningfully past this particular limitation, a point section 9

returns to directly as a recommended direction for future work.

8.3 The one dimension that clearly worked, and why

Pressing intensity is the exception to nearly everything described in section 8.2, and there is a specific, football grounded reason it carries real signal where most of the other dimensions mostly do not. Pressing intensity measures a defensive, out of possession behaviour that is comparatively decoupled from a team's own technical or possession quality. A team does not need to be technically excellent in order to press intensely, and a technically excellent team does not automatically press intensely as a result of that excellence alone. This reasoning, however, is only partly borne out by what this project's own final manager profiles actually show, and the part that does not match the popular narrative turns out to be genuinely instructive rather than merely disappointing.

De Zerbi's centroid does carry the lowest pressing intensity score of the three managers, exactly as this project's own qualitative research predicted well before any quantitative modelling began, and this is independently consistent with the specific qualitative finding that his signature tactical mechanism operates while his own team has the ball, deliberately baiting the opponent into pressing him so that his own team can play directly through that press, rather than something that happens defensively after his team has already lost the ball.

But the popular narrative's more specific prediction, that Klopp's teams would score highest of the three on this dimension, does not hold up in the final, corrected data. Guardiola's centroid, at zero point three zero four, sits marginally higher than Klopp's centroid, at zero point two two nine, on the final, possession adjusted measure, with both of these scores sitting clearly above De Zerbi's centroid, at negative zero point zero zero one. The most likely explanation for this specific reversal, and one that is entirely consistent with this project's own central diagnosis described in section 8.2, is that the possession adjustment applied to pressing statistics, described in section 6.5, corrects specifically for how much possession a team's opponents typically have, not for squad quality in any more general sense. Manchester City's possession dominance inside this project's own sample is so extreme, even by the standards of other elite clubs, that their opponents genuinely have unusually little of the ball, which plausibly over corrects Manchester City's own adjusted pressing rate upward, beyond what pure managerial pressing intent alone would justify. This is best understood not as a failure of the adjustment itself, but as a concrete, specific illustration of exactly the same residual confound this project has described in general terms throughout sections 8.2 and elsewhere. Seeing that confound appear specifically, inside the one dimension this whole project relies on most heavily, is a useful and honestly reported data point in its own right, not something to minimise or explain away.

What survives cleanly, and what this project can state with real confidence, is that pressing intensity separates De Zerbi from the other two managers exactly as predicted in advance, and that it remains, by a wide margin, the single most discriminating tactical dimension in this project's entire feature set, according to the permutation importance analysis described in section 6.5. That is a real, statistically supported result, simply a more precisely qualified one than an earlier, less careful

version of this same report first described it as.

9. Conclusiones y Trabajo Futuro

This project demonstrates a complete, reproducible, methodologically disciplined pipeline for knowledge informed manager philosophy modelling, together with an honest evaluation of exactly where that pipeline's current limits lie. The research question stated in section three can now be answered directly. Football knowledge and player performance data can genuinely be combined into an explainable compatibility framework, the architecture works exactly as designed, and every single score this project produces is traceable back to named, football language evidence, never a black box number. But the specific feature set built inside this project's four and a half working weeks, spread across four sprints, does not yet fully separate three real managers' footballing philosophies from the effects of squad and league resourcing strength, with one clear, credible, statistically confirmed exception, pressing intensity, which is itself independently corroborated by qualitative tactical research conducted before any quantitative modelling began. This is a materially more defensible conclusion than either overclaiming a clean success that the evidence does not actually support, or abandoning the project's central question as unanswerable, and it is offered here explicitly as such.

Future work is recommended here in priority order, reflecting a deliberate choice made by the project owner to accept and report the current finding honestly rather than attempt a full redesign inside this project's remaining timeline.

The single most promising direction for future work is to explicitly account for a player's own team strength when computing his tactical dimension scores, comparing a player not only against the wider league population in his position, as this project already does, but additionally against his own teammates specifically, a within team contrast alongside the current between league contrast. This would directly target the confound diagnosed in section 8.2, rather than working around it indirectly the way the current possession adjustment and role based grouping fixes do.

A second promising direction is to move beyond aggregated match level statistics toward positional tracking data, which would make it possible to build genuinely structural features, the shape formed by outfield players during build up play, or the specific sequence and tempo of a pressing trigger, rather than only the countable outcomes of that structure. This would require a different, considerably more restricted data source than the one this project used, but it would directly address the fundamental limitation identified throughout section 8.2.

A third, more modest direction, achievable relatively quickly if this project were to continue, is to close the one clearly identified remaining gap in this project's own qualitative research, a thinner than desired investigation into De Zerbi's specific out of possession defensive tactics, which would strengthen the football grounded interpretation of his profile considerably.

A fourth direction concerns extending historical coverage for Bayern Munich and Borussia Dortmund, but only if a genuinely new data source is found in the future. The active search this project already conducted, described in section 6.2, should be treated as exhausted for now, a closed line

of investigation rather than a standing open task.

A fifth direction, using the starting formation recorded for every team in every match, has already been explored during this project and is described fully in section 6.6. The formation data itself held up well once recovered, and the resulting descriptive profiles are genuinely informative, independently corroborating this report's qualitative description of De Zerbi as the most tactically fluid of the three managers. It was not folded into the compatibility score as a full eighth dimension, because formation data is only available for the twenty two club seasons inside this project's own manager attributed sample, far too small a population to standardise safely alongside the other seven dimensions, each of which draws on the full eleven thousand plus player season population. Closing that remaining gap, so that formation could genuinely be added as a proper eighth dimension rather than a descriptive supplement, would require either locating a data source with formation coverage across the full player population, or an explicit statistical treatment suited specifically to a small sample, and remains a genuine, well scoped direction for future work rather than something this project could complete inside its own timeline.

Finally, and this is stated as a firm recommendation rather than a suggestion, no version of this system as it currently stands should be presented to a football club, or to any other real world decision maker, as a validated recruitment tool. Even at the improved forty two point one percent best fit rate, this is meaningfully but not overwhelmingly above chance, and a tool built directly on today's compatibility scores should continue to be presented exactly as it is presented inside the application itself, as a research demonstration with its limitations stated clearly and prominently, not as validated recruitment guidance, until further work of the kind described above has been carried out and independently evaluated.

10. Bibliografia

This section gives the full, consolidated list of references this project relied on, organised into three groups exactly as they were actually used during the work, rather than as a single undifferentiated list.

Academic literature. Seven academic papers on football playing style modelling were read in full and synthesised into roughly twenty two thousand eight hundred words of notes underlying the ontology described in section 6.2. The single most important of the seven, referenced repeatedly throughout sections six and eight, is Plakias, Armatas, Mprotsis, Kyranoudis, Gu, Xylomenos, and Michailidis, published in two thousand twenty five in the Journal of Physical Education and Sport under the title "What playing styles best represent the overall tactical identity of European soccer teams", which provides the statistically validated nineteen dimension taxonomy this project adopted as the anchor for its own seven dimension feature set. The full reference list, in full academic citation form, is as follows.

Gu, Linda, Wang, Wenlong, Plakias, Spyridon, and Zhang, Shaoliang. Playing style identification in team sports, a systematic review from statistical dimensionality reduction to unsupervised machine learning. International Journal of Sports Science and Coaching, two thousand twenty five.

Plakias, Spyridon, Armatas, Vasilis, Mprotsis, Theodoros, Kyranoudis, Angelos, Gu, Linda, Xylomenos, Fotios, and Michailidis, Yiannis. What playing styles best represent the overall tactical identity of European soccer teams. Journal of Physical Education and Sport, volume twenty five, number eleven, two thousand twenty five, pages two thousand two hundred forty nine to two thousand two hundred fifty seven.

Zhong, Yonghan, Xu, Ying, Zhu, Kecheng, Diaz Cidoncha Garcia, Jorge, Gomez Ruano, Miguel Angel, and Yi, Qing. Data driven classification of playing styles and match outcome prediction in UEFA Champions League teams. Biology of Sport, volume forty three, two thousand twenty six, pages five hundred seventy five to five hundred eighty six.

Bolin, Niklas, and Vesterlind, Elliot. From Patterns to Playbooks, identifying offensive and defensive team playing styles from possession chains with machine learning and its possible practical applications in management of football teams. Master's thesis, Uppsala University, Faculty of Science and Technology, June two thousand twenty five.

Tang, Zitian, Wang, Xing, and Zhang, Shaoliang. Clustering Football Game Situations via Deep Representation Learning.

Baattite, Anass. Machine Learning Based Football Tactic and Style Analysis. Capstone design final report, School of Science and Engineering, Al Akhawayn University, two thousand twenty three.

Wang, Zhe, Velickovic, Petar, Hennes, Daniel, Tomasev, Nenad, Prince, Laurel, Kaisers, Michael, Bachrach, Yoram, Elie, Romuald, Wenliang, Li Kevin, Piccinini, Federico, Spearman, William, Graham, Ian, Connor, Jerome, Yang, Yi, Recasens, Adria, Khan, Mina, Beauguerlange, Nathalie, Sprechmann, Pablo, Moreno, Pol, Heess, Nicolas, Bowling, Michael, Hassabis, Demis, and Tuyls, Karl. TacticAI, an AI assistant for football tactics. October two thousand twenty three, arXiv preprint two thousand three hundred ten point one zero five five three.

Three further sources were identified as cited by the papers above but not read in full for this project, and are flagged here rather than silently omitted, since an honest bibliography should distinguish what was actually read from what was only encountered secondhand. Cintia and Pappalardo's Coach2vec, which models a coach's own style directly rather than a team's, and which any future extension of this project modelling managers rather than teams should read before making any claim of novelty. Plakias, Kokkotis, and colleagues' earlier two thousand twenty three paper, the original source of the nineteen latent variables that the two thousand twenty five paper above reuses. And Rein and Memmert's two thousand sixteen paper, cited elsewhere as the strongest general argument for why a football knowledge layer, rather than raw statistics alone, is needed at all.

Manager tenure and tactical philosophy sources. Seventy seven individually attributed sources were used to build and cross check the manager tenure table described in section 6.2 and the qualitative tactical research described throughout sections three and eight, each one corroborated against at least two independent sources per specific claim. These include club official statements,

Wikipedia's own managerial statistics tables, analyst pieces from Opta and StatsPerform, Coaches' Voice, Spielverlagerung, Total Football Analysis, and named press outlets. A representative sample of the most heavily used sources follows. Marti Perarnau's Pep Confidential, accessed via Perarnau Magazine, for Guardiola's own positional play framework. Coaches' Voice pieces titled Pep Guardiola Coach Watch, The tactical evolution of Pep Guardiola's Manchester City, and Roberto De Zerbi tactics and style of play. Spielverlagerung's Manchester City positional play analysis and its preview of Guardiola's Manchester City reign. Multiple data confirmed statistical pieces on Guardiola's Manchester City tenure from Opta Analyst. SoccerTutor and a Medium analysis by SHfootball on De Zerbi's goalkeeper and centre back build up mechanism. And Wikipedia's own managerial statistics tables for all three managers, used for tenure dates and cross corroborated against club and press sources with every conflict explicitly resolved and documented.

Data source documentation. StatsBomb Open Data, a public repository of competition and match manifests and event level data. The worldfootballR_data mirror maintained by JaseZiv, the primary source of FBref and Opta match level advanced statistics used throughout this project, described in full in section 6.2. The soccerdata Python package, tested for direct FBref and Understat access and found insufficient on its own, described in section 6.2. And Hugo Mathien's European Soccer Database on Kaggle, investigated as a possible extension to cover Bayern Munich and Borussia Dortmund's earlier eras and found unsuitable, also described in section 6.2.

11. Anexos

This section provides four appendices in full, rather than only pointing to the files they are drawn from, so that a reader can consult the underlying evidence directly inside this report itself.

Appendix A, further screenshots of the application

Section 6.6 already shows three screenshots of the application working through the Erling Haaland example in full. Two further tabs of the same application are shown here for completeness.

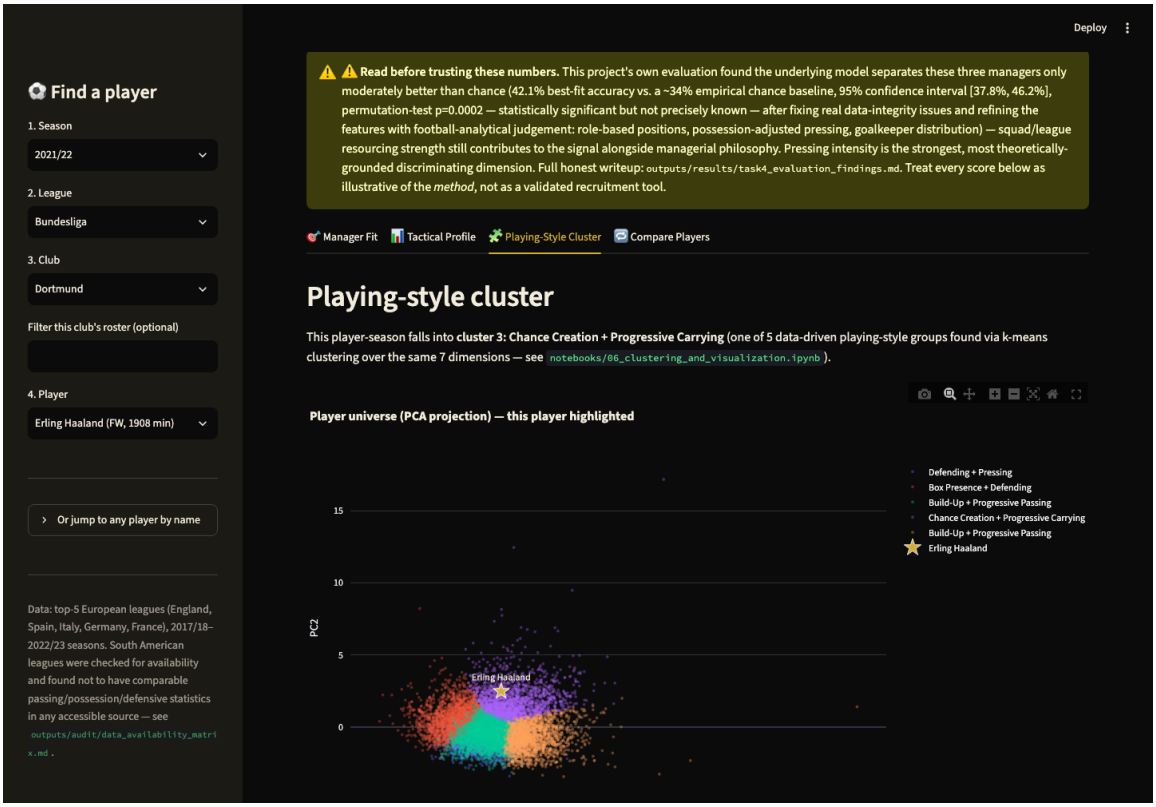


Figure 8: The Playing Style Cluster tab, screenshotted directly from the application, showing Erling Haaland's star marker sitting at the boundary between two clusters within the full population of eleven thousand one hundred and thirty two player seasons.

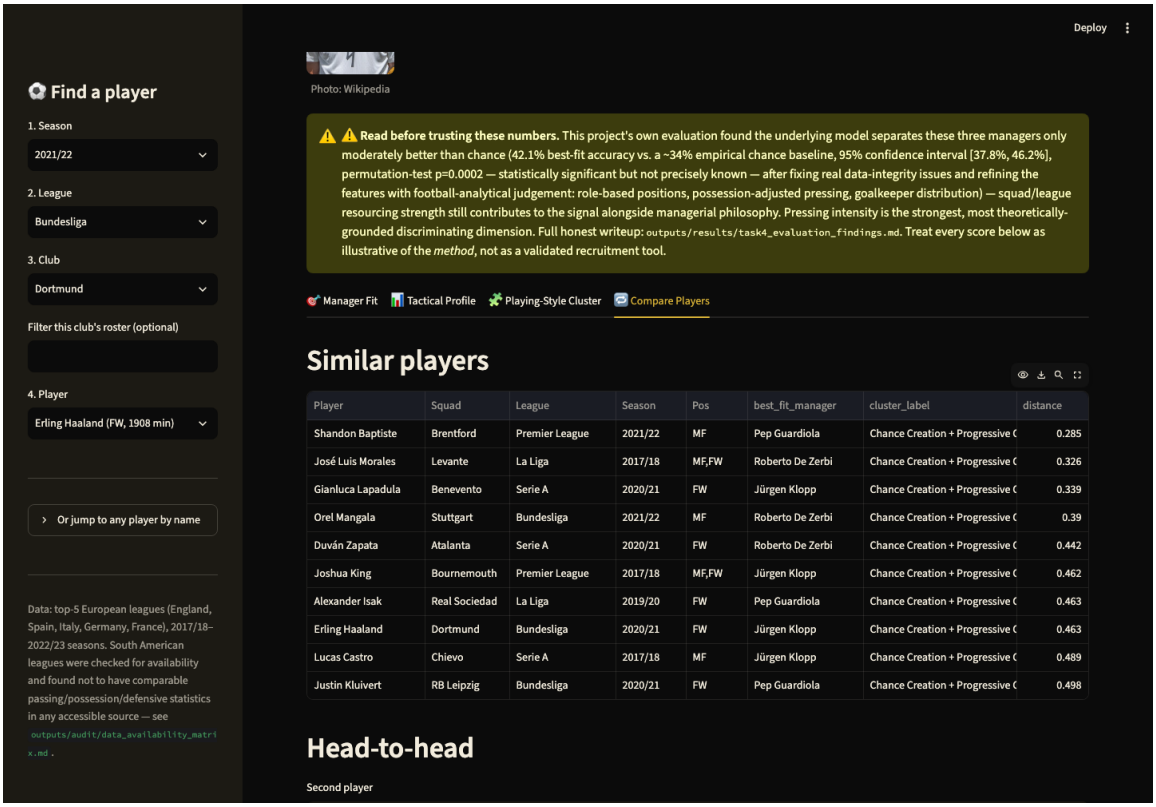


Figure 9: The Compare Players tab, screenshotted directly from the application, showing the head to head radar chart comparison tool available for any two players in the full population.

Appendix B, the verified manager tenure table in full

The table below is the complete, twenty row record described in sections 6.1 through 6.3, covering every club and national team role held by the three managers across their careers, whether or not that spell was ultimately included in the quantitative sample. A safety flag of safe means the entire spell can be attributed at season level without risk. Caution means the spell is a whole season but the sample itself is compromised in some other way, for instance a semi professional league with no usable statistics, or a league season that was abandoned partway through. Unsafe means the manager changed partway through that specific season, so any use of that season must split at the exact match date rather than attribute it wholesale. Exclude means the row is not a club coaching role at all, for instance a national team post with no matches played yet, and was never a candidate for the quantitative sample.

Row	Manager	Club	Seasons covered	Safety
G1	Pep Guardiola	FC Barcelona B	2007/08	Caution
G2	Pep Guardiola	FC Barcelona	2008/09 to 2011/12	Safe

Row	Manager	Club	Seasons covered	Safety
G3	Pep Guardiola	Bayern Munich	2013/14 to 2015/16	Safe
G4	Pep Guardiola	Manchester City	2016/17 to 2025/26	Safe
GN1	Pep Guardiola	National team, none held	not applicable	Exclude
K1	Jurgen Klopp	1. FSV Mainz 05	2000/01 part, to 2007/08	Unsafe, 2000/01
K2	Jurgen Klopp	Borussia Dortmund	2008/09 to 2014/15	Safe
K3	Jurgen Klopp	Liverpool	2015/16 part, to 2023/24	Unsafe, 2015/16
K4	Jurgen Klopp	Red Bull, non coaching role	not applicable	Exclude
KN1	Jurgen Klopp	Germany national team	2026/27 onward	Exclude, no matches yet
D1	Roberto De Zerbi	US Darfo Boario	2013/14 part	Unsafe, and no data
D2	Roberto De Zerbi	Foggia Calcio	2014/15 to 2015/16	Safe
D3	Roberto De Zerbi	US Citta di Palermo	2016/17 part	Unsafe
D4	Roberto De Zerbi	Benevento Calcio	2017/18 part, from matchday 10	Unsafe
D5	Roberto De Zerbi	US Sassuolo Calcio	2018/19 to 2020/21	Safe
D6	Roberto De Zerbi	FC Shakhtar Donetsk	2021/22, abandoned season	Caution
D7	Roberto De Zerbi	Brighton and Hove Albion	2022/23 part, to 2023/24	Unsafe, 2022/23
D8	Roberto De Zerbi	Olympique de Marseille	2024/25 to 2025/26 part	Unsafe, 2025/26
D9	Roberto De Zerbi	Tottenham Hotspur	2025/26 part, final 7 fixtures, to 2026/27	Unsafe, not used as a style sample
DN1	Roberto De Zerbi	National team, none held	not applicable	Exclude

Appendix C, the ten concepts and seven tactical dimensions in summary

Section 6.4 explains in detail how these were built and why. The table below gives a compact reference for what each one actually measures.

Concept	What it measures
Progressive passing	The share of a player's passes that move the ball meaningfully closer to the opponent's goal
Progressive carrying	The share of a player's ball carries that move the ball meaningfully closer to the opponent's goal
Possession retention	How reliably a player keeps the ball once he has it, relative to his position
Chance creation	A player's contribution to creating scoring opportunities for others
Box presence	How often a player is present and involved inside the opponent's penalty area
Pressing intensity	How aggressively a player and his team contest the ball out of possession, adjusted for how much of the ball their opponents actually had
Individual defending	A player's own duels, tackles, and defensive actions relative to his position
Aerial play	A player's involvement and success in aerial duels
Crossing	A player's crossing volume and end product
Goalkeeper distribution	How far and how directly a goalkeeper distributes the ball, added in the analyst driven refinement described in section 6.5

Tactical dimension	Built from	Football meaning
Buildup directness	Progressive passing, possession retention, goalkeeper distribution	How directly a team progresses the ball from the back
Progression passing	Progressive passing alone	Passing based progression specifically
Progression carrying	Progressive carrying alone	Ball carrying based progression specifically
Chance creation	Chance creation, crossing	Overall creative output
Box presence	Box presence alone	Penalty area occupation

Tactical dimension	Built from	Football meaning
Pressing intensity	Pressing intensity alone	Out of possession defensive aggression, the strongest dimension in the whole feature set
Individual defending	Individual defending, aerial play	Duel and aerial defensive contribution

Appendix D, the project's decision log in full

The following ten entries record every material methodological decision made across this project, in the order they were made, together with the alternatives considered and the evidence behind each final choice. They are reproduced here in full because they are, collectively, the most direct evidence of this project's own methodological discipline, referenced throughout sections six and eight above.

One. CRISP DM was adopted as the mandatory methodological framework for the entire project, not as one option among several, because both guideline documents for this programme require it explicitly and require a dedicated video explaining how it was applied. This shaped both the notebook structure and this report's own structure, organised by CRISP DM phase throughout section six.

Two. FBref and Opta match level statistics, accessed through the public worldfootballR_data mirror, were chosen as the primary data source over scraping FBref directly, because direct scraping was tested and blocked by the source's own Cloudflare protection from this project's working environment, with no local browser available to work around it. This trades a fixed historical coverage window, described in section 6.2, for reliable, reproducible access.

Three. FBref and Opta match level data and StatsBomb Open Data are never pooled into a single feature matrix without an explicit harmonisation step, because FBref itself changed its underlying data provider from StatsBomb to Opta in late October twenty twenty two, and the two providers are not guaranteed to define even identically named statistics the same way. Guardiola's Barcelona era, only available through StatsBomb, is therefore treated throughout as an independent qualitative check, never blended into the main quantitative manager profile built from the Opta era.

Four. Manager attribution is always done at match level using the verified tenure table, never at season level, because eight of the twenty tenure rows in that table are marked unsafe for season level attribution, including two of the richest data seasons otherwise available. This is the single rule described in section 6.1 as the most consequential business understanding decision in the whole project.

Five. Guardiola's Bayern Munich, Klopp's Borussia Dortmund, and De Zerbi's Palermo and Shakhtar Donetsk spells were excluded from the quantitative feature set after an explicit, actively

conducted search for an adequate alternative data source came up empty, described in full in section 6.2. Benevento's two thousand seventeen to eighteen season was added back in as a corrected inclusion partway through the project, once full match level data for that season was found to exist after all, though it remains flagged as a low confidence sample given known squad quality issues.

Six. This is the most consequential entry in the whole log. FBref's match level passing, passing types, possession, and goalkeeper statistical blocks turned out to be entirely unpopulated for Serie A and Ligue 1 before the two thousand twenty three to twenty four season, a gap that had been present since the very first data collection step but only surfaced once the club identity bias check described in section 6.5 produced a suspiciously clean split between clubs. Investigating that suspicious result, rather than accepting it, surfaced this gap together with a second bug, a line of code silently converting missing data into a literal zero rather than leaving it as genuinely unknown. Both were fixed. No aggregation step anywhere in this project's code is now allowed to treat missing data as zero, every function was rewritten to skip missing values and renormalise its weights accordingly, and every similarity comparison now records exactly how many of the seven dimensions it actually had data for. Even after both fixes, the first honest result was thirty two point eight percent best fit accuracy, statistically no better than chance, a genuine and sobering finding, not a residual bug, reported honestly rather than hidden.

Seven. Following a deliberate review of the whole pipeline from the perspective of a football analyst rather than an engineer, three concrete weaknesses were identified and fixed together, described in full in section 6.5, an eight way role based position grouping replacing the earlier four coarse buckets, a possession adjustment applied to pressing statistics, and a new goalkeeper distribution concept built from data that had already been collected but never used. Two further problems were caught and fixed while implementing these three, not originally planned. The dimension scoring function needed to be made aware of missing values at the individual row level, or the new goalkeeper concept would have silently broken the buildup directness dimension for every outfield player. And the first attempt at the new position grouping was itself briefly broken, dumping the vast majority of players into a single bucket, because match level data uses a more specific set of position tags than season level data does, caught by inspecting the actual resulting distribution rather than assuming the new code worked correctly. Implementing all three fixes together raised the best fit accuracy to forty two point one percent, described and explained at length in section 6.5 and section 8.

Eight. Two further additions were made once the forty two point one percent figure was established. A formal bootstrap confidence interval and permutation test were added around that figure, described in section 6.5, since a single point estimate on its own does not say how precisely it is known. And each match's starting formation, collected since early in the project but never analysed, was explored as a possible eighth tactical dimension. Checking the live data source before building this addition revealed that it is a continuously updated release rather than a version specific archive, and had already moved past the historical window this project needs, described in full in section 6.6. The historical formation data was recovered from an earlier saved intermediate file

rather than lost, and the resulting formation profiles are reported descriptively, kept separate from the numerical compatibility score, because the population they are drawn from is far too small to standardise safely alongside the other seven dimensions.

Nine. A systematic fact checking pass, carried out specifically to assess whether this project was ready to submit, compared every specific numeric claim already written into this report against the actual saved output files, rather than trusting the already written text. Two real errors were found and corrected this way. The formation population size had been stated as nine club seasons in several places when the real figure, verified directly against the saved data, is twenty two club seasons across six clubs. And the permutation importance figures reported after the sprint seven fixes were materially wrong, understating pressing intensity's true importance and incorrectly describing box presence as having become a positive contributor when it had not. Both were corrected throughout this report, and this entry itself is the record of that correction, in keeping with this project's own stated discipline of checking a written claim against its source rather than assuming it correct because it was written carefully.

Ten. A further, more extensive verification pass checked every qualitative comparative claim about the three managers' tactical profiles directly against the saved manager profile data and the already published bar chart figure, rather than trusting the popular narrative the earlier text had repeated. This project's own earlier drafts had stated that Klopp's manager profile scores higher than Guardiola's on pressing intensity, consistent with his well known reputation. The actual, verified figures say the opposite, Guardiola's centroid sits marginally higher than Klopp's on the final, possession adjusted measure, though De Zerbi's centroid remains clearly the lowest of the three exactly as predicted. This is explained at length in section 8.3, together with the most likely reason for the reversal, and every section of this report that had repeated the earlier, incorrect version of this claim was corrected to state the true result honestly.